

Pr. 3

Part A - Theoretical Foundation.

1) what is Statistical Distribution?

→ ~~Statistical distribution~~ ~~Kisi datadi ke bar~~
A statistical distribution shows how data values are spread and how often they occur.

2) what is a Q-Q plot and why is it used?

→ A Q-Q plot compares sample data with a ~~theoretical~~ theoretical distribution to check if the data follows the distribution.

3) Difference b/w Discrete and Continuous Distribution

- Discrete: Countable values (e.g. number of transactions)
- Continuous: Any value within a range (e.g. transaction amount)

4) what is Bernoulli Distribution

→ A probability distribution with only two outcomes: Success (1) or Failure (0).

5) what is Binomial Distribution?

→ A distribution that measures the number of successes in a fixed number of independent trials.

6) Explain log-Normal Distribution.

A distribution where the logarithm of the variable follows a normal distribution. It is commonly used for positive, right skewed data.

7) Explain Power Law Distribution.

→ A distribution where the small values occur frequently and very large values occur rarely.

8) What is Box-Cox Transform?

→ A mathematical transformation used to reduce skewness and make data closer to normal.

9) Explain Poisson Distribution with an Example.

→ Poisson distribution models the number of events occurring in a fixed period.

Ex: Number of transaction per day.

10) What is Z-Score probability?

→ A Z-score measures how many standard deviations a value is from the mean and helps calculate probabilities.

11) Difference Between PDF and CDF.

- PDF (Probability Density at a specific value)
- Used for Continuous data.

PDF

CDF

- Shows the cumulative probability up to a given value.
- Always increases from 0 to 1.